COURSE PLAN

UPES/Course Plan/Theory & lab/Ver 10/ Effective December 2025

Name of School

Name of the Faculty	
Designation	
Cluster	
Program	B.Tech (CSE)
Course	Object Oriented Programming
Course Code	CSEG1044
No. of credits	04 (L-03 and P-01)
Semester	4th
Session	Jan – May, 2026
Academic Year	2025-2026

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Signature of Faculty Member

Name: _____

Designation: _____

Date: _____

_____

Signature of Cluster Head

Name: _____

Designation: _____

Date: _____

COURSE PLAN

Prerequisite	**Programming in C**	
Credit	**04**	
Lecture	**Tutorial**	**Practical**
03	**0**	**01**

A. The expected Program Outcome are:

PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9	Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

B. Expected Program specific Outcome are:

PSO1	Perform system and application programming using computer system concepts, the principles of data structures, algorithm development, problem-solving, and optimization techniques.
PSO2	Apply software development and project management methodologies, integrating principles from both front-end and back-end development, and effectively utilize contemporary tools and technologies.
PSO3	Exhibit a commitment to ethical practices, societal responsibilities, and continuous learning, contributing to the advancement of technology and addressing challenges in diverse computing domains.

C. The expected Course Outcomes are:

THEORY	
CO 1	Understand Object Oriented Programming concepts and architecture of Java.
CO 2	Analyze and model the real-world entity using Java programming language
CO 3	Develop packages with Generics and Implement Interfaces with Exception handling.
CO 4	Create Stand-alone Java applications using GUI swings and JDBC.

LAB	
CO 1	Understand Object Oriented Programming concepts and architecture of Java.
CO 2	Analyze and model the real-world entity using Java programming language
CO 3	Develop packages with Generics and Implement Interfaces with Exception handling.
CO 4	Create Stand-alone Java applications using GUI swings and JDBC.

D. CO-PO Relationship Matrix

Indicate the relationships by1- Slight (low) 2- Moderate (Medium) 3-Substantial (high)

Program Outcome / Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	1	-	-	-	-	-	-	-	-	-	-	-	2	3	-
CO 2	-	3	3	-	-	-	-	-	-	-	-	-	2	3	-
CO 3	-	3	3	-	-	-	-	-	-	-	-	-	2	3	-
CO 4	-	-	-	2	-	-	1	-	2	2	-	-	2	3	-
Average	1	3	3	2	-	-	1	-	2	2	-	-	2	3	-

E. Course Outcomes assessment plan:

a) Mapping of Course Outcomes with assessment tools

Components / Course Outcomes	Assignment	Test/Quiz	Mid Semester	End Semester	Any other (Project/Viva)
CO1	☑	☑	☑	☑	☑
CO2	☑	☑	☑	☑	☑
CO3	☑	☑	☑	☑	☑
CO4	☐	☑	☐	☑	☑

b) Assessment plan and Analysis: THEORY

	Assessment Tools	Planned	Actual	Highest (Score)	Lowest (Score)
CIA- Theory (25%)	Assignment (10 marks)				
	Quiz 1 (10 marks)				
	Quiz 2 (10 marks)				
	Class Test (20 marks)				
	Project/Viva (50 marks)				
Mid Term Exam (20%)					
End Term Exam (30%)					

c) Assessment plan and Analysis: Lab

	Assessment Tools	Planned	Actual	Highest (Score)	Lowest (Score)
CIA- Lab (25%)	Quiz-1 (10 marks)				
	Quiz-2 (10 marks)				
	Lab Test-1 (15 marks)				
	Lab Test-2 (15 marks)				
	Continuous Lab Assessment (CLA) (50 marks)				

F. Course Syllabus Template:

Syllabus Template for Theory

Unit Number	Content
Unit I	**Introduction to OOPs** Object Oriented Programming History and Evolution, Object Oriented Programming Principles, Features of Java, Input Output Statements, Comment Line Arguments, Data Types, Variables, Operators, Program Control Statements, Arrays, Type of Arrays, Strings. (CO1)
Unit II	**Classes, Inheritance, Packages and Interfaces** Class Fundamentals, Objects, Constructors, Garbage Collection, this Keyword, Java's Access Modifiers, Method Overloading, static Keyword, Inheritance, Types of Inheritance, super to Access Superclass Members, Method Overriding, Abstract Classes, Using final, Packages and Interfaces, Build-in Interface, User defined Interfaces. (CO1, CO2)

Unit III	**Nested Classes, Exceptions, Multithreading & IO Streams** Nested Classes, Types of Nested Classes, Exception Handling, Exception Handlers, Concurrent Programming, The Thread Class and Runnable Interface, Thread Priorities, Synchronization, Java's I/O Streams, Byte Streams and Character Streams, FileWriter, FileReader. (CO2, CO3)
Unit IV	**Generics, Lambdas, GUI Swing & Database Connectivity** Generics Fundamentals, Generic Class, Generic Methods, Lambdas, Functional Interfaces, Swing, Components and Containers, Layout Managers, Swing Event Handling, Event Listeners, Event Classes and Listener Interfaces, Swing Controls, Database Connectivity, Statement, Prepared Statement, CallableStatement, Resultset. Persistent Data. (CO3, CO4)
Unit V	**Collections and Wrapper Class** Collections, Iteration, Collection Interface, Set and SortedSet, List, Map and SortedMap, Wrapped Collections and Collections Class, Wrapper classes and loading classes. (CO3)
Unit VI	**Capstone Project** Create Standalone Java Project, Designing of UML and database diagrams, GUI Panel development using swing, Establish connection with Database and Panel. Source Code Management and Collaboration using Git/GitHub. Unit Testing using JUnit, Integration Testing, Build and Artifactory Management. (CO4)

Syllabus Template for lab

Course Title: Object Oriented Programming – CSEG1044

It is designed to provide hands-on experience in Java programming with a strong emphasis on object-oriented principles. The experiments are structured to allow flexibility to instructors while ensuring uniform learning outcomes across all sections.

GENERAL INSTRUCTIONS

- Students must write and execute programs independently.
- Programs should follow standard Java coding conventions.
- Each experiment submission must include source code, output, and brief observations.
- Faculty members are free to define specific problem scenarios within the scope of the experiment.

UNIT I: INTRODUCTION TO OOP & JAVA FUNDAMENTALS

Experiment 1: Introduction to Java Programming Environment

Objective: To understand the Java programming environment, program structure, compilation, and execution process.

CO Mapping: CO1

Problem Statement: Design and implement a basic Java program demonstrating class declaration, main() method, and standard output statements.

Tasks to be Performed: - Install and configure JDK - Write a basic Java program - Compile and execute the program - Observe compilation and runtime behavior

Constraints / Guidelines: - Follow Java syntax and standards - Use meaningful identifiers

Expected Outcome: Students will be able to write, compile, and execute a basic Java program.

Experiment 2: Input Handling and Command Line Arguments

Objective: To understand different methods of input handling in Java.

CO Mapping: CO1

Problem Statement: Develop a Java program that accepts input using standard input methods and command line arguments.

Tasks to be Performed: - Use Scanner or BufferedReader - Process command line arguments - Display formatted output

Expected Outcome: Students will be able to handle input from multiple sources.

Experiment 3: Data Types, Variables, and Operators

Objective: To apply data types and operators in Java programs.

CO Mapping: CO1

Problem Statement: Write a Java program demonstrating the use of primitive data types and operators.

Expected Outcome: Students will understand data manipulation using operators.

Experiment 4: Decision Making and Looping Constructs

Objective: To implement control flow using conditional and looping statements.

CO Mapping: CO1

Problem Statement: Design a Java program using decision-making and looping constructs.

Expected Outcome: Students will be able to control program flow effectively.

Experiment 5: Arrays and Multidimensional Arrays

Objective: To work with one-dimensional and two-dimensional arrays.

CO Mapping: CO1

Problem Statement: Develop Java programs to perform operations on arrays.

Expected Outcome: Students will be able to store and process data using arrays.

Experiment 6: String Handling

Objective: To perform string manipulation using Java String classes.

CO Mapping: CO1

Problem Statement: Implement string operations using built-in Java methods.

Expected Outcome: Students will be able to manipulate textual data.

UNIT II: CLASSES, INHERITANCE, PACKAGES & INTERFACES

Experiment 7: Class Design and Object Creation

Objective: To understand classes, objects, and constructors.

CO Mapping: CO1, CO2

Problem Statement: Design classes with data members and member functions.

Expected Outcome: Students will understand object creation and encapsulation.

Experiment 8: Method Overloading and Constructors

Objective: To demonstrate compile-time polymorphism.

CO Mapping: CO2

Problem Statement: Implement method and constructor overloading in Java.

Expected Outcome: Students will understand method resolution at compile time.

Experiment 9: Inheritance and Method Overriding

Objective: To implement inheritance and runtime polymorphism.

CO Mapping: CO2

Problem Statement: Create base and derived classes demonstrating inheritance.

Expected Outcome: Students will understand code reusability.

Experiment 10: Abstract Classes and Interfaces

Objective: To implement abstraction using abstract classes and interfaces.

CO Mapping: CO2

Problem Statement: Design programs using abstract classes and interfaces.

Expected Outcome: Students will understand abstraction and multiple inheritance.

UNIT III: EXCEPTIONS, MULTITHREADING & I/O

Experiment 11: Exception Handling

Objective: To handle runtime errors using exception handling mechanisms.

CO Mapping: CO2, CO3

Problem Statement: Develop programs using try-catch-finally blocks.

Expected Outcome: Students will write robust programs.

Experiment 12: Multithreading

Objective: To implement multithreading in Java.

CO Mapping: CO3

Problem Statement: Create threads using Thread class and Runnable interface.

Expected Outcome: Students will understand concurrent programming.

Experiment 13: File Handling Using Java I/O Streams

Objective: To perform file operations using Java I/O streams.

CO Mapping: CO3

Problem Statement: Develop programs using FileReader and FileWriter.

Expected Outcome: Students will handle persistent data.

UNIT IV: GENERICS, LAMBDAS, GUI & JDBC

Experiment 14: Generics

Objective: To implement generic classes and methods.

CO Mapping: CO3

Problem Statement: Design programs using generics.

Expected Outcome: Students will understand type safety.

Experiment 15: Lambda Expressions

Objective: To use lambda expressions and functional interfaces.

CO Mapping: CO3

Problem Statement: Implement lambda expressions in Java programs.

Expected Outcome: Students will understand functional programming.

Experiment 16: GUI Programming Using Swing

Objective: To design GUI applications using Swing.

CO Mapping: CO4

Problem Statement: Develop GUI applications using Swing components.

Expected Outcome: Students will create interactive applications.

Experiment 17: Database Connectivity Using JDBC

Objective: To connect Java applications with databases.

CO Mapping: CO4

Problem Statement: Implement JDBC connectivity using Statement and PreparedStatement.

Expected Outcome: Students will interact with databases.

UNIT V: COLLECTIONS & WRAPPER CLASSES

Experiment 18: Collections Framework

Objective: To use Java Collection interfaces.

CO Mapping: CO3

Problem Statement: Develop programs using List, Set, and Map.

Expected Outcome: Students will manage collections of data.

Experiment 19: Wrapper Classes

Objective: To understand wrapper classes and autoboxing.

CO Mapping: CO3

Problem Statement: Implement programs using wrapper classes.

Expected Outcome: Students will understand primitive-object conversion.

UNIT VI: CAPSTONE PROJECT

Experiment 20: Mini Project

Objective: To integrate OOP concepts into a standalone Java application.

CO Mapping: CO4

Problem Statement: Design and implement a Java mini project with GUI and database support.

Expected Outcome: Students will develop a complete Java application.

Faculty Instruction Note

- Faculty members must adhere to the experiment titles and objectives.
- Specific problem statements may be customized.
- Evaluation should be based on correctness, design, and explanation.

G. Target

Target %	50
Level 1	40
Level 2	50
Level 3	60

H. Method of Evaluation*

UG/PG (Indicative)	
Quizzes/Tests/Assignments/Class Test/ Viva/Project/CLA/Lab Tests (50%)	
Mid Examination (20%)	
End examination (30%)	

*It can be revised as per the assessment scheme of the respective School/Course & In case of the courses running under continuous assessment this scheme is not applicable.

I. Passing Criteria

Scale	PG	UG
Out of 10 point scale	SGPA – "6.00" in each semester CGPA – "6.00" Min. Individual Course Grade – "C" Course Grade Point – "4.0"	SGPA – "5.0" in each semester CGPA – "5.0" Min. Individual Course Grade – "C" Course Grade Point – "4.0"

*for PG, passing marks are 40/100 in a paper (Composite)

*for UG, passing marks are 35/100 in a paper (Composite)

J. References:

Textbooks	1. Herbert Schildt, *Java: A Beginner's Guide*, 10th Edition**,** McGraw-Hill Education, 2024. 2. Herbert Schildt, *Java: The Complete Reference***,** 12th Edition, McGraw-Hill Education, 2024.
Reference books	1. Joshua Bloch, *Effective Java*, **3rd Edition**, Addison-Wesley, 2018. 2. Allen B. Downey and Chris Mayfield, *Think Java: How to Think Like a Computer Scientist*, 2nd Edition**,** O'Reilly Media, 2020.
Web Resources	1. Oracle Java Documentation https://docs.oracle.com/javase/ 2. Java Tutorials by Oracle https://docs.oracle.com/javase/tutorial/
Journals	
MOOCs, online courses	1. Programming with Java – SWAYAM https://onlinecourses.swayam2.ac.in 2. Java Programming – NPTEL (IITs) https://nptel.ac.in/courses/106